

③ Combinational - Assignment

1. Implement the following functions using
 i) 2:1 mux and NOT gates
 ii) 4:1 Mux.

a) $F(A, B, C) = \bar{A}C + B\bar{C}$

b) $F(P, Q, R) = \bar{P}\bar{R} + Q\bar{R} + \bar{P}Q$

= a) i) Choose C as select line
 for 2:1 mux $F = \bar{S}T_0 + ST_1$

Here $S = C$

When $C = 0$ $f = B$ $\therefore T_0 = B$

When $C = 1$ $f = \bar{A}$ $\therefore T_1 = \bar{A}$

Connections

• Select = C • $T_0 = B$ • $T_1 = \bar{A}$ (using one NOT gate)

ii) Take A and B as select lines

A	B	F
0	0	C
0	1	1
1	0	0
1	1	$\bar{C}$

Hence • $S_1 = A$ • $S_0 = B$

Inputs are $T_0 = C$ $T_1 = 1$ $T_2 = 0$ $T_3 = \bar{C}$

b) i) Choose R as select

when $R=0$

$$F = \bar{P} + Q$$

$$\therefore \bar{P} + Q = \begin{cases} 1, & P=0 \\ Q, & P=1 \end{cases}$$

when $R=1$

$$F = \bar{P}Q$$

$$\therefore \bar{P}Q = \begin{cases} Q, & P=0 \\ 0, & P=1 \end{cases}$$

$$\therefore \underline{T_0} = \bar{P} + Q$$

$$\underline{T_1} = \bar{P}Q$$

ii) Take P and Q as select lines

P Q F

0 0 $\bar{R}$

0 1 1

1 0 Q

1 1 R

$$\therefore S_1 = P$$

$$S_0 = Q$$

1/ps

$$\underline{T_0 = \bar{R}, T_1 = 1, T_2 = 0, T_3 = \bar{R}}$$

2. Implement the functionality of a full adder using 2:1 muxes only.

Sum :- $S = A \oplus B \oplus C_{in}$

⇒ Choose A as select

when $A = 0$,

$$S = B \oplus C_{in}$$

when $A = 1$,

$$S = \overline{B \oplus C_{in}}$$

∴ Select = A

$$T_0 = B \oplus C_{in}$$

$$T_1 = \overline{B \oplus C_{in}}$$

} The XOR itself is implemented using 2:1 mux.

Choose B as select.

$$∴ T_0 = C_{in}$$

$$T_1 = \overline{C_{in}}$$

Carry :-

$$C_{out} = AB + AC_{in} + BC_{in}$$

Choose A as select

when $A = 0$,

$$C_{out} = BC_{in}$$

when $A = 1$,

$$C_{out} = B + C_{in}$$

Hence,

$$T_0 = BC_{in}$$

$$T_1 = B + C_{in}$$

3. Simplify $f(a, b, c, d) = \sum m(2, 4, 5, 6, 10) +$
 $D(12, 13, 14, 15)$ and implement the SOP expression
 using 4:1 Mux.

$$f(a, b, c, d) = \sum m(2, 4, 5, 6, 10) + D(12, 13, 14, 15)$$

$$\therefore f = ab + \bar{a}(b + c)$$

$$f = ab + b\bar{a} + c\bar{a}$$

a b F using 4:1 mux

$$0 \quad 0 \quad c\bar{a}$$

$$0 \quad 1 \quad 1$$

$$1 \quad 0 \quad c\bar{a}$$

$$1 \quad 1 \quad 1$$

$$\therefore S_1 = a$$

$$S_0 = b \quad \therefore T_0 = c\bar{a} \quad T_1 = 1$$

$$T_2 = c\bar{a} \quad T_3 = 1$$

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